CHAPTER 7

PHOTOSYNTHESIS IN HIGHER PLANTS

CHAPTER ESSENCE AT A GLANCE:
Solar radiation drives photochemical thylakoid reactions generating ATP and NADPH to power enzymatic carbon fixation in the stroma.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Photochemical Z-Scheme and Water Splitting:** Light reactions occur in thylakoid membranes via Photosystem II (P680) and Photosystem I (P700). Non-cyclic photophosphorylation drives electrons via an electron transport chain from water to NADP+. Water photolysis occurs on the lumen side of PS II (releasing O2, 4H+, and 4e-), while cyclic photophosphorylation in stroma lamellae involves only PS I, generating ATP alone.

• **Chemiosmotic ATP Generation Mechanism:** Peter Mitchell Chemiosmotic hypothesis explains ATP synthesis by a trans-thylakoid proton gradient: protons accumulate within the thylakoid lumen due to water photolysis, plastoquinone proton translocation, and stroma NADP+ reduction. Proton efflux through CF0 channels drives conformational rotation in the CF1 headpiece of ATP synthase, synthesizing ATP from ADP and inorganic phosphate.

• **The Calvin C3 Cycle Stoichiometry:** The Calvin cycle in the chloroplast stroma comprises three enzymatic phases: (1) Carboxylation: CO2 combines with ribulose 1,5-bisphosphate (RuBP) catalyzed by RuBisCO to form 3-phosphoglycerate (3-PGA); (2) Reduction: 2 ATP and 2 NADPH are consumed per CO2 fixed to form triose phosphate; (3) Regeneration: 1 ATP is consumed to regenerate RuBP. Synthesizing 1 glucose molecule requires fixing 6 CO2 molecules, consuming 18 ATP and 12 NADPH.

• **Kranz Anatomy and C4 Photosynthetic Adaptation:** C4 plants (maize, sugarcane) overcome photorespiration through Kranz anatomy: mesophyll cells lack RuBisCO and utilize PEP carboxylase to fix CO2 into oxaloacetic acid (OAA); malate transports to bundle sheath cells, releasing CO2 around RuBisCO at high partial pressures, eliminating oxygenase activity and maximizing photosynthetic biomass yield.

ESSENTIAL TAXONOMICAL FRAMEWORKS & PROCESSES

$$6CO_2 + 12H_2O \xrightarrow[\text{Chlorophyll}]{\text{Light}} C_6H_{12}O_6 + 6H_2O + 6O_2$$

Overall Photosynthetic Transduction Reaction

$$2H_2O \xrightarrow[\text{PS II (P680)}]{\text{Oxygen Evolving Complex}} 4H^+ + 4e^- + O_2\uparrow$$

Photolysis of Water in the Thylakoid Lumen

$$\text{Calvin Cycle Cost} = 6CO_2 + 18ATP + 12NADPH \longrightarrow \text{Glucose} + 18ADP + 12NADP^+$$

Stoichiometric Energetics per Molecule of Synthesized Glucose

$$\text{C}_4\text{ Primary Carboxylation} = PEP + HCO_3^- \xrightarrow{\text{PEP Carboxylase}} \text{Oxaloacetate (4C)}$$

Atmospheric CO2 Trapping in C4 Mesophyll Cells

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** For every single CO2 molecule fixed in the C3 cycle, exactly 3 ATP and 2 NADPH are consumed; to generate 1 glucose molecule (6 carbons), exactly 18 ATP and 12 NADPH are required.

• **Alert:** In C4 plants, an additional 2 ATP per CO2 are consumed for PEP regeneration, making the total energetic cost 30 ATP and 12 NADPH per glucose.

• **Alert:** Photorespiration (C2 cycle) occurs when RuBisCO binds oxygen instead of CO2 under high oxygen and high temperature, wasting energy without producing ATP or sugar.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. State Blackman's Law of Limiting Factors. Name any two internal (plant) factors that affect the rate of photosynthesis. [Score: 2 Marks]

Answer & Key Valuation Steps:

• Blackman's Law of Limiting Factors (1905): If a chemical process is affected by more than one factor, its rate is determined by the factor that is nearest to its minimal value; it directly affects the process if its quantity is changed. **[1 Mark]**

• Internal plant factors: Number, size, age, and orientation of leaves; mesophyll cells and chloroplasts; internal CO_2 concentration; amount of chlorophyll. **[1 Mark]**

2. Distinguish between cyclic and non-cyclic photophosphorylation on the basis of photosystems involved, photolysis of water, and end products. [Score: 2 Marks]

Answer & Key Valuation Steps:

• Photosystems: Non-cyclic involves both PS II and PS I; cyclic involves only PS I. **[0.75 Mark]**

• Photolysis of water: Occurs in non-cyclic with oxygen evolution; absent in cyclic with no oxygen release. **[0.75 Mark]**

• End products: Non-cyclic yields both ATP and NADPH; cyclic yields only ATP without NADPH. **[0.5 Mark]**

3. Explain the Chemiosmotic hypothesis of ATP generation in chloroplasts, detailing the three causes of proton accumulation inside the thylakoid lumen. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Chemiosmotic hypothesis: ATP synthesis is linked to the development of a proton concentration gradient across the thylakoid membrane. **[0.75 Mark]**

• Water splitting: Photolysis of water occurs on the inner side of the thylakoid membrane, releasing protons (H^+) directly into the thylakoid lumen. **[0.75 Mark]**

• Proton translocation: Plastoquinone transfers electrons from PS II to cytochrome b6f and pumps protons from the stroma into the lumen. **[0.75 Mark]**

• NADP reduction: NADP reductase on the stromal side removes protons from the stroma to form NADPH, depleting stromal protons and steepening the gradient; proton exit through CF0-CF1 drives ATP synthesis. **[0.75 Mark]**

4. Describe the three major enzymatic stages of the Calvin C3 cycle, specifying the energetic cost in terms of ATP and NADPH per glucose molecule synthesized. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Carboxylation: Fixation of atmospheric CO_2 into a stable 3-carbon compound 3-phosphoglycerate (3-PGA) using ribulose 1,5-bisphosphate (RuBP), catalyzed by RuBisCO. **[0.75 Mark]**

• Reduction: Series of reactions leading to glucose formation; utilizes 2 ATP for phosphorylation and 2 NADPH for reduction per CO_2 molecule fixed. **[0.75 Mark]**

• Regeneration: Regeneration of the CO_2 acceptor RuBP requires 1 ATP per turn of the cycle. **[0.75 Mark]**

• Total energetic cost: For 1 molecule of glucose (6 CO_2 fixed), 18 ATP (12 in reduction + 6 in regeneration) and 12 NADPH are consumed. **[0.75 Mark]**

5. What is Kranz anatomy? Explain how the spatial division of labour between mesophyll and bundle sheath cells enables C4 plants to prevent photorespiration. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Kranz anatomy refers to the wreath-like arrangement of large, radially arranged bundle sheath cells around vascular bundles with several layers, thick walls, and no intercellular spaces. **[1 Mark]**

• Mesophyll cells contain PEP carboxylase which has a high affinity for CO_2 and fixes it into 4-carbon oxaloacetate, lacking RuBisCO. **[1 Mark]**

• Malate transports to bundle sheath cells and decarboxylates, releasing high concentrations of CO_2 around RuBisCO, ensuring it functions purely as a carboxylase and preventing oxygenation (photorespiration). **[1 Mark]**

6. Why is photorespiration considered a wasteful process in C3 plants? Name the three organelles involved in the photorespiratory cycle. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Wasteful process: Under high oxygen and temperature, RuBisCO binds O2 instead of CO2, converting RuBP into 3-PGA and phosphoglycolate without producing ATP, NADPH, or sugar, while releasing previously fixed CO2 with ATP expenditure. **[1.5 Marks]**

• The three cooperating organelles in the C2 photorespiratory cycle are: (1) Chloroplast, (2) Peroxisome, and (3) Mitochondrion. **[1.5 Marks]**

7. Calculate the total number of ATP and NADPH molecules required for the synthesis of one molecule of glucose in a C3 plant and a C4 plant respectively. Explain why the C4 plant requires extra ATP. [Score: 4 Marks]

Answer & Key Valuation Steps:

• C3 plant energetic cost: Fixing 6 molecules of CO2 requires 6 turns of the Calvin cycle, consuming 18 ATP and 12 NADPH per glucose molecule synthesized. **[1.5 Marks]**

• C4 plant energetic cost: Requires 18 ATP and 12 NADPH for the Calvin cycle in bundle sheath cells, plus 12 additional ATP for regenerating PEP from pyruvate in mesophyll cells, totaling 30 ATP and 12 NADPH per glucose molecule. **[1.5 Marks]**

• Justification for extra ATP: The additional 2 ATP per CO2 fixed (12 ATP per glucose) represent the biological energetic investment to pump CO2 and eliminate photorespiratory carbon loss, yielding net higher biomass under hot arid conditions. **[1 Mark]**